

“I Want It That Way”: Exploring Users’ Customization and Personalization Preferences for Home Assistants

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ABSTRACT

Home assistants are becoming a widespread product, but they mostly come as a compact device and offer very few customization and personalization features, which often leads to dissatisfaction. With the technological advancements, these systems are becoming more adaptable to the users’ needs and can better imitate a human personality. To achieve that efficiently, understanding how different users envision their desired assistant is crucial. To identify people’s customization and personalization preferences and their desired personality for a home assistant, we designed a set of storyboards depicting a variety of possible features in a domestic setting and conducted a user study ($N = 15$), including a series of semi-structured interviews. Our quantitative results suggest that users prefer an agent which is highly agreeable and has higher conscientiousness and emotional stability. Furthermore, we discuss users’ customization and personalization preferences for a home assistant, which could be considered when designing the future generation of home assistants.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**; *User studies*.

KEYWORDS

Voice Assistants, Home Assistants, Personality, Big Five, Customization, Personalization

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1 INTRODUCTION

The use of voice assistants (VAs) is spreading widely and home assistants such as Apple HomePod, Google Home, or Amazon Echo are becoming increasingly common [10]. VAs are designed to provide assistance whenever they are called upon for various different tasks such as smart home control [26, 38, 46], scheduling [50, 59], navigation [29, 51], education [5, 57], and entertainment [4, 62].

Despite all the benefits that such systems provide, it still remains a difficult task to design a fulfilling experience for users as the interaction with voice-controlled systems is often characterized as troublesome [46], disappointing [37], or embarrassing [16]. This is partially due to limitations in speech recognition and partially because these systems do not fulfill the user’s expectations [32, 37, 41, 63]. Due to the individual differences in preferences for a desired home assistant, systematic adaptation of these systems to the user is challenging [55]. Therefore, commonly available devices tend to take a one-size-fits-all approach, which ignores the possible advantages of adapting to user preferences. Moreover, previous research has shown that users unconsciously assign personalities to conversational agents [48], and purposefully manipulating this perception can affect users’ trust and engagement with such devices [14, 64].

Customization and personalization features have been shown to enhance user satisfaction and improve performance [18, 20, 21, 40, 42]. Customized features are those the user can explicitly select between specific options, whereas personalized features are driven by computers based on users’ individual needs [44].

In this work, we present a human-centered approach [11] intending to identify customization and personalization preferences of home assistants, as well as desired personality types of the agent. More specifically, this work is guided by the following research questions:

- RQ1: How do users imagine the personality of their desired home assistant?
- RQ2: In which ways do users want to customize and personalize their home assistants?

We conducted semi-structured interviews using storyboards that contained scenarios of everyday domestic situations. Our results

show that users prefer an agent with high *agreeableness*, *conscientiousness*, and *emotional stability*. We also discuss users' customization and personalization preferences for a home assistant and their implications for future research avenues of personalized assistants.

2 RELATED WORK

There has been extensive research regarding home assistants and how people interact with such devices [8, 19, 36]. In this section, we discuss research on voice assistant customization, personalization, and the importance of personality in human-agent interaction.

2.1 VA Customization

Customization describes the extent to which technology or service can be modified to comply with user preferences [31, 49]. In contrast, *personalization* refers to automatic adaptation to users' needs based on observed behaviors [31]. Providing users with control over interaction can improve performance and user satisfaction [58]. A large body of research has highlighted the importance of customization in VAs, particularly for people with special needs [1, 2, 40, 42, 58]. Molnar and Kletke [40] emphasize that the lack of flexibility reduces productivity and satisfaction. Murad et al. [42] identified lack of control as a common cause of user frustration and highlighted the need for user control and freedom in speech interfaces. Therefore, to enhance the experience, VAs should allow for the configuration of speed, tone, and volume along with other characteristics of the virtual agents [2]. Furthermore, Zargham et al. [61] observed users' desire to customize additional design factors of home assistants beyond voice characteristics, such as the number of agents and their roles and personalities. Moreover, allowing users to customize the agent's appearance to their liking could potentially improve the interaction with such systems [10].

2.2 VA Personalization

Asif and Krogstie [7] define personalization as "a controlled process of adaptation of a service to achieve a particular goal by utilizing the user model and the context of use." The user model refers to the recorded data, which includes user information with the aim to adapt systems to the individual needs [52]. User models can be created with the user's own entries or systems automatically adapting to user behavior, which is also referred to as direct and indirect user input [39]. However, while direct user inputs might be very advantageous in the first place, an inappropriately set assistant may be less accepted by a user than its default configuration [12]. Therefore, many systems additionally apply indirect adaptation of user behavior [39, 66]. Personalization yields a number of benefits to business owners and customers since it allows for higher efficiency and user acceptance for a product or service [30, 45]. Studies have shown that users prefer VAs that can adapt to their preferences and background [12, 20–22, 34]. People tend to find it easier to interact with technologies that partially resemble their own characteristics [13]. A study by Braun et al. [12] has shown that users prefer and trust a personalized in-car VA more than the default version, especially when the agent's personality matches the user's.

2.3 Personality and VAs

Many innovative technologies, including home assistants, tend to be seen as social actors in general [43], which describes how users apply social rules and assign them personalities [48] while interacting with such systems. An extensive body of work has examined the role of personality in human-computer interaction [9, 24, 25, 27, 47, 48]. The Five-Factor Model (FFM), also known as the big five [33] is the most widely accepted personality theory in scientific research for modeling human personality [24, 25, 53] which boils down human personality to five core factors of *Openness*, *Conscientiousness*, *Extraversion*, *Agreeableness*, and *Neuroticism*.

Personality has been described as an important aspect for designing and adapting voice assistants [47]. While the personality of the user has an impact on their desired agent, designing the agent's personality is also crucial. Research has shown that distinctive agent personalities could be successful in promoting trust and user acceptance [6, 9, 27, 60]. For instance, Chen et al. [17] suggest that users prefer an extroverted agent over an introverted one and are more talkative towards it. Moreover, the authors found that agents' personality traits influence users' preferences, dialogues, and behavior. Andrist et al. [6] showed that matching the agent's personality to the user's has a positive effect on users' motivation and willingness to interact with the agent. For chatbots, Völkel and Kaya [54] found that users with higher agreeableness prefer an agreeable agent. Understanding user perceptions and preferences for the personality of the agent can improve user experience [35].

Eventually, customization and personalization features of voice user interfaces (VUIs) have been shown to enhance the user experience with these devices. Moreover, designing specific personalities for agents could also impact the experience. Nevertheless, there is still a gap in terms of understanding users' preferences for such agent features.

To build upon the prior work, we engage users to think about possible features of home assistants to shed further light on how to build an ideal agent. We specifically focus on users' customization and personalization preferences for home assistants and how they envision the personality of their desired agent with respect to their own personality.

3 STUDY DESIGN

We conducted a user study to explore users' customization and personalization preferences, as well as their desired personalities for home assistants. For this, we used an approach inspired by scenario-based design methods [15] and vignette experiments [3], where we present a series of hypothetical situations to the participants and ask them to reflect on the scenarios. This approach is similar to the so-called Speed Dating method [23, 65] and allows investigating technologies despite current technical limitations. Using this method, designers can identify possible problems as well as opportunities regarding specific technologies, and create more appropriate and innovative solutions [23]. Moreover, a variety of use cases can be considered which do not necessarily apply to the participant but are relevant for other users, e.g., impairments, medical conditions, or social contacts. We used graphical storyboards to visualize the situational context and spatial relation between the characters and the assistant.

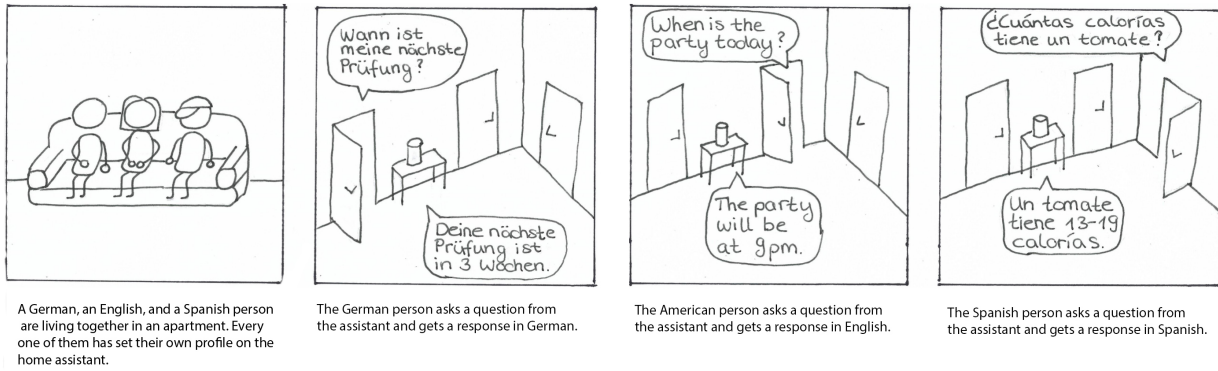


Figure 1: Exemplary storyboard that was used in the interview in which the home assistant responds differently to each household member.

Scenarios and Storyboards. Iteratively and in discussions, we developed scenarios within the context of the home environment involving either a single person or multiple people. First, in brainstorming sessions between two researchers, 24 scenarios were conceived covering different aspects in a domestic setting. The scenarios were based on what researchers imagined being a compelling customization/personalization feature of home assistants in everyday situations which they were familiar with themselves, or they knew from others. After three iterations, we selected ten scenarios from the initial list covering different features, made final adjustments, and created the storyboards for the chosen scenarios. We classified the scenarios based on the type of assistance and the number of people involved. For instance, in the scenario shown in Figure 1, the agent's type of assistance is fact-checking, and the people involved are the three people who live in the same apartment, making it a multi-party scenario. Using these storyboards, we ran a pilot study with two participants to find possible flaws within the study design and to see if the scenarios inspired participants to reflect on. We worked with different styles of storyboards in an iterative process with the aim to minimize the cultural and ethnic cues so that the participants can put themselves in the shoes of the characters. The appearance of the home assistants was similar to a common cylinder shaped smart speaker available on the market. Furthermore, the characters within the storyboards did not have face expressions in an attempt to avoid any influence from the reactions of the characters. An exemplary scenario with the final design is depicted in Figure 1. The complete set of storyboards designed and used for this research can be found in the supplementary materials. In the following, we briefly describe the scenarios:

- *S1 Multilingual:* The assistant can understand multiple languages simultaneously and can respond accordingly.
- *S2 Pace Customization:* The user adjusts the speaking pace of the assistant to ease the use for an elderly user.
- *S3 Speaking Adaptation:* The assistant picks up the speaking characteristics of the user and responds similarly.
- *S4 Limited Access:* The user limits the interaction with the assistant in a way that it only responds to the owner.
- *S5 Interest Adaptation:* The assistant learns about the users' interests and uses that when responding to inquiries.

- *S6 Mood Adaptation:* The assistant senses the users' current mood and adjusts its responses based on that.
- *S7 Multi-Persona:* The assistant is set with multiple characters with distinct personalities and areas of responsibility.
- *S8 Self Embodiment:* The user sets their embodied agent to look like them.
- *S9 Shared Assistant:* Two siblings use the same personalized agent that is only capable of having one user profile.
- *S10 Private Information:* The user sets the assistant to share certain information only with the owner.

Procedure. The interview sessions were held remotely via video calls, where the experimenter recorded verbal statements and observations while providing assistance in cases of issues. The screen of the interviewer was shared to display the storyboards. In the beginning of the session, the participants were briefed about the procedure of the experiment. After giving informed consent, all participants filled out a demographics questionnaire and stated their prior experience with home assistants as well as their typical usage of VAs. Then, the participants stated their personality traits using the Ten Item Personality Inventory (TIPI) [28]. TIPI is a 10-item questionnaire that assesses personality according to the Five-Factor Model. Each item is rated on a 7-point Likert scale ranging from 1 (disagree strongly) to 7 (agree strongly). After this, the semi-structured interviews started where in random order, the participants were presented with all ten scenarios one by one and commented on each story. First, participants were asked to give their impressions about the individual scenarios, and point out the positive and negative aspects of the interaction with the home assistant. Afterwards, they provided overall suggestions and recommendations in terms of customization and personalization, as well as the desired personality of a home assistant. Finally, the participants filled in a second TIPI where they stated the desired personality of the voice agent. For further, qualitative analysis, we recorded the audio of each session. A single session took between 50-75 minutes.

Participants. 15 participants at the age between 22 and 41 ($M = 26.6$, $SD = 4.29$) volunteered for our study (4 self-identified as female, 11

male). All participants had previous experience with home assistants (4 rarely, 11 often), 12 owned a smart speaker. We conducted the experiment in the local official language to avoid language barriers.

4 RESULTS

We performed a thematic content analysis to evaluate patterns within participants' responses. The transcripts of the interviews were independently coded by two researchers using inductive coding, where a single quote could be assigned to multiple codes. The researchers organized, reviewed, and discussed the codes, resolved disagreements, and derived themes which were categorized into (i) *agent's speech characteristics*, (ii) *agent's visualization*, (iii) *agent's personality*, and (iv) *Privacy and Security*. In this section, we first present the analysis of the personality questionnaires and then discuss the interview findings.

Personality. To investigate the self-personality differs significantly from the desired personality of the home assistant, we performed paired-sample t-tests on each subscale of the TIPI between both responses. The descriptive statistics and the results of the t-tests are shown in Table 1. The analysis identified significant differences for agreeableness, conscientiousness, and emotional stability between the self-reported personality and the desired personality.

To determine if the desired personality of the home assistant is in line with the self-personality of the users, we performed correlation analysis between the two ratings on all subscales of TIPI. The analysis showed several significant correlations. A positive correlation was found between the agent's *agreeableness* and the users' *agreeableness* with a Pearson's coefficient of $r = .550$ ($p = .033$). Further, the agent's *emotional stability* negatively correlated with users' *openness to experiences* with $r = -.556$ ($p = .031$).

Interview Analysis. Overall, participants demanded more customization and personalization features for a home assistant. All but one participant mentioned that the personalization features of the current home assistants are not sufficient. 13 participants would be willing to pay extra for a system that can be more personalized. 12 people mentioned that they would use a better customized home assistant more often. The qualitative analysis of the interviews identified four overarching themes: *agent's speech characteristics*, *agent's visualization*, *agent's personality*, and *Privacy and Security*. We report the results of the interview based on the categories.

Agent's Speech Characteristics. Participants found it highly useful to be able to modify the speed, tone, and volume of the agent's voice, and participants demanded to have the ability to adjust the voice themselves. However, users had concerns regarding the difficulty of making such adjustments ("is it adjustable by my grandmother?"). Eight participants said that the agent's voice should sound more human-like and were not satisfied with their current agent's voice. Eleven participants preferred a specific gender for their desired agent's voice. Participants also said that their desired agent's voice should sometimes convey certain emotions such as excitement, sarcasm, friendliness, and calmness. On the other hand, two participants demanded a neutral voice which doesn't convey any specific emotions. Six Participants mentioned that celebrity voices could

be used for the agent as an entertaining feature. One user found this as "momentary entertainment", while one participant mentioned "I would not like to use a voice that I know from anywhere, it makes me uncomfortable". All participants mentioned that the assistant has to understand different languages to be able to assist more people, and three participants suggested it could encourage users to learn foreign languages. Five users mentioned changing the language from the settings is not sufficient and this should be done automatically.

Agent's Visualization. Six participants preferred the agent to have a digital embodiment, where the other nine did not find it helpful. From those who preferred an embodied agent, all wanted a humanoid representation, and two participants mentioned that the visual representation should match the voice. Two participants mentioned that the agent should look healthy and fit. Three participants specifically asked for a female embodiment for the agent. For most users, the humanoid clothing wasn't that important. However, two users preferred the assistant to occasionally change its outfit. Regarding the agent's presence on the screen, six participants preferred that the agent is constantly present. Users said comments such as "If I see the assistant, it makes me want to talk to it" or "I think it's a good reminder that the assistant is there, sometimes I may forget I have one if it is not speaking". However, the other nine participants preferred the assistant to appear only when asked. One user said, "I would feel observed if it's constantly there".

Personality Adaptation. 12 participants were in favor of agent's adapting to the user's personality. One participant said, "it should adjust the information content based on my desires, not necessarily behave like me". Another suggested "people change, so the agent should be able to change and keep up with the changes in my interest". Eight participants were in favor of mood adjustments by the agent, where the others were resistant against this feature. Participants in favor mentioned that "it makes the device seem more human". On the other hand, users also said "I don't need a piece of software to show me empathy, I know it's programmed". One user had doubts regarding the VA's judgment "If it misjudges my mood, it would work horribly wrong", while another participant feared that "this might make people rely on such devices and have less interaction with real humans". Ten participants wanted the agent to use humor, while two suggested that the agent should be serious and prevent using humor.

Privacy and Security. Eleven people mentioned that they would trust the VA more if it provides more security and privacy related features. All participants were concerned regarding the agent sharing personal information in front of others. They all agreeably mentioned that an access control is a helpful feature that home assistants could provide, meaning that the home assistant identifies the role of the users and responds accordingly. A participant mentioned that this could be very helpful for families with small children. One user stated that "the system should be able to respond appropriately to the users that are not the owner". Another said, "It should talk to friends and family the same as to me, but talk more seriously to my colleagues". One user suggested that "The agent should distinguish between age groups, talk differently to kids than to adults". Participants had different suggestions regarding sharing

Table 1: Descriptive statistics and results of paired-sample t-tests (df=14) of TIPI ratings for self-personality and desired personality of the VA.

	Self-Personality M (SD)	Agent-Personality M (SD)	<i>t</i> (14)	<i>p</i> -value	Cohen's <i>d</i>
Extroversion	4.13 (1.18)	4.63 (1.07)	1.3693	0.1925	0.443
Agreeableness	4.70 (1.42)	5.56 (0.96)	2.792	0.0144	0.712
Conscientiousness	5.20 (1.39)	6.63 (0.71)	3.930	< 0.01	1.285
Emotional Stability	4.80 (1.08)	6.20 (0.64)	3.7936	< 0.01	1.568
Openness to Experiences	5.40 (1.22)	5.40 (1.27)	0.0	1.0	0

private information. One believed that the system should include standard and private features. Another said that "private information should require a fingerprint/password". One user proposed that "the system should have specific modes where the microphone is deactivated for a specific time or in specific rooms".

5 DISCUSSION

The results from our scenario-based study suggest that participants are not satisfied with the customization and personalization aspects of current home assistants and demand more of such features.

Regarding the personality of the agent (RQ1), users rated equally or higher in all the five scales of the TIPI for their desired agent's personality in comparison to their own personality ratings. We found significantly higher ratings in terms of *agreeableness*. Furthermore, for this subscale, a strong positive correlation between the users' and their desired agent's was observed. This finding further supports previous work by Völkel and Kaya [54] where they found that users with higher agreeableness prefer an agreeable agent. Participants also rated significantly higher in subscales of *conscientiousness* and *emotional stability* for the agent in comparison to self-personality. These subscales constitute reliability, which is an important factor in human-agent communication. It is not surprising that users expect such systems to behave reliably, since they have to trust the agent with their tasks. This personality trait was even more crucial for users who rated themselves lower on *openness to experiences*. This finding is supported by the negative correlation between the agent's *emotional stability* and the users' *openness to experiences*.

Concerning RQ2, the qualitative analysis revealed interesting tendencies towards different customization and personalization features. In line with previous work [1, 18], being able to modify an agent's voice characteristics such as gender, speed, tone, and volume was highly favored by the participants. Most users preferred to control and adjust such characteristics themselves, rather than it being a personalization feature. However, some raised concerns about the difficulty of such adjustments and demanded a simple user interface. Commonly, the customization features of the current home assistants, require users to use a graphical interface such as their smartphone to configure the VA. On many occasions, users have difficulties to find the dedicated section in the app for specific features. Further, in homes with multiple household members, if the person whose smartphone is connected to the device is not available, such modifications would not be possible. These obstacles could frustrate users and make the customization process rather complex.

A number of our participants who owned a smart speaker requested features that are already available in their systems. This underlines that users are often not aware of the existing customization and personalization features.

In line with Andrist et al. [6], participants were mostly in favor of the agent matching the user's personality and adapting to their interests, and users would highly benefit from such personalized systems. However, users expected the agent to evolve with any changes in the users' interests. Although most participants wanted the assistant to show emotions, be more humorous, sense users' mood and adjust the responses to it, and overall behave more human-like, some were still rather hesitant towards such features. The main reason against the human-like behavior of the agent was the mistrust in the technology. Users believed if the agent falsely recognizes users' mood and responds inappropriately, it would affect the users extremely negatively. Another concern raised was that some users did not like to see human-like features in an agent and considered such interactions as fake. These users wanted to see a clear distinction between a human and a software. Such agent capabilities could be attractive personalization features to consider when designing the future home assistants. Nevertheless, since such social skills are complex abilities that are still very difficult for computer systems to master, we recommend providing customizable features in which the users are in charge of setting when and how the agent should use such skills.

Features which could potentially improve accessibility, such as supporting multiple languages simultaneously, were considered to be highly valuable by the participants. A number of users proposed creative and fun ideas such as having the agent sound or look like a celebrity such as musicians, actors, or athletes. Six participants also showed interest in having a digital embodiment for the agent, all of which preferred a humanoid. Modifying further details regarding the agent's representation such as hairstyle, clothing and body size was also requested by our participants. Moreover, some users wanted the agent to occasionally change its visualization, suggesting the agent has a daily life. However, similar to Bonfert et al. [10], participants raised concerns regarding the constant presence of an embodied agent. Consequently, since we learned that users' preferences regarding agent embodiment often vary and no universal solution could satisfy the expectations of all users equally. Therefore, we recommend having the digital embodiment as a customizable feature.

Furthermore, we observed that privacy concerns are still one of the main challenges of home assistants. Users want to have more

privacy related features in order to have greater trust and to avoid inconvenient situations. Features such as having individual user profiles, user roles, and access control were suggested to better protect users' privacy. Beyond that, the participants wanted to have different modes for particular circumstances where the agent varies the language depending on the persons in the room to avoid inconvenient or inappropriate situations. Future home assistants would presumably require collecting more personal data from users in order to provide better services. Therefore, we encourage developers to consider such privacy features in the design of future generations of these systems.

5.1 Limitations and Future Work

Although the findings of this work present important considerations for designing home assistants, Our research and the findings are still limited in several ways which need to be addressed. Firstly, in this study, we used a limited set of scenarios covering only a specific range of possible features. As a result, our participants may not fully understand the range of services or possibilities for customization and personalization features of a home assistant. In order to scale up the insights of our work, future research can expand these scenarios by investigating other possible features. Moreover, since many of the features used in the storyboards are not available in the market, we explored people's perceptions of such features by presenting hypothetical scenarios, meaning that the participants did not actually experience the situation. One can assume that participants' feelings and perceptions about these features could differ once they interact with them in real life. Therefore, future work should investigate the user experience in the scenarios while interacting with the home assistant. The experiment sample mostly consisted of male users (11 out of 15). An influence of such bias on the results cannot be excluded. Hence, future studies should validate our findings by investigating a wider population. We will expand our experiment design and investigate closer how different demographic factors such as gender and cultural background corroborate our results. Lastly, a recent study suggests that the big five model may not be applicable to describe the personality of a conversational agent, since this model was derived from human language use in order to describe human personality [56]. However, in our work, we consider a concept of future VA technology that aims to adapt human-like behavior. Therefore, while agreeing with Völkel et al. that the FFM may not be suitable to assess the current state-of-the-art VAs, we argue that it is the most eligible method to describe human-like behavior of VAs. In our future studies, we plan to develop a home assistant which conveys the desired agent personality based on the present results and evaluate it in terms of usability and UX.

6 CONCLUSION

In this work, we explored users' customization and personalization preferences for home assistants as well as the desired personality of the agent. We created ten scenarios which were depicted in storyboards that demonstrate how VAs might act in different domestic situations and interviewed 15 participants. The results show that users prefer an agent which is highly agreeable, and has high conscientiousness and emotional stability. Furthermore,

the interviews yielded four categories of features users wish for home assistants (*agent's speech characteristics*, *agent's visualization*, *agent's personality*, and *Privacy and Security*). These findings highlight that current assistants contain several limitations and provide important suggestions and considerations which could guide the design of future generation home assistants in order to achieve a higher user satisfaction.

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